

*Differences between
Istanbul, Türkiye
&
Alphen a/d Rijn, NL
in 2050*

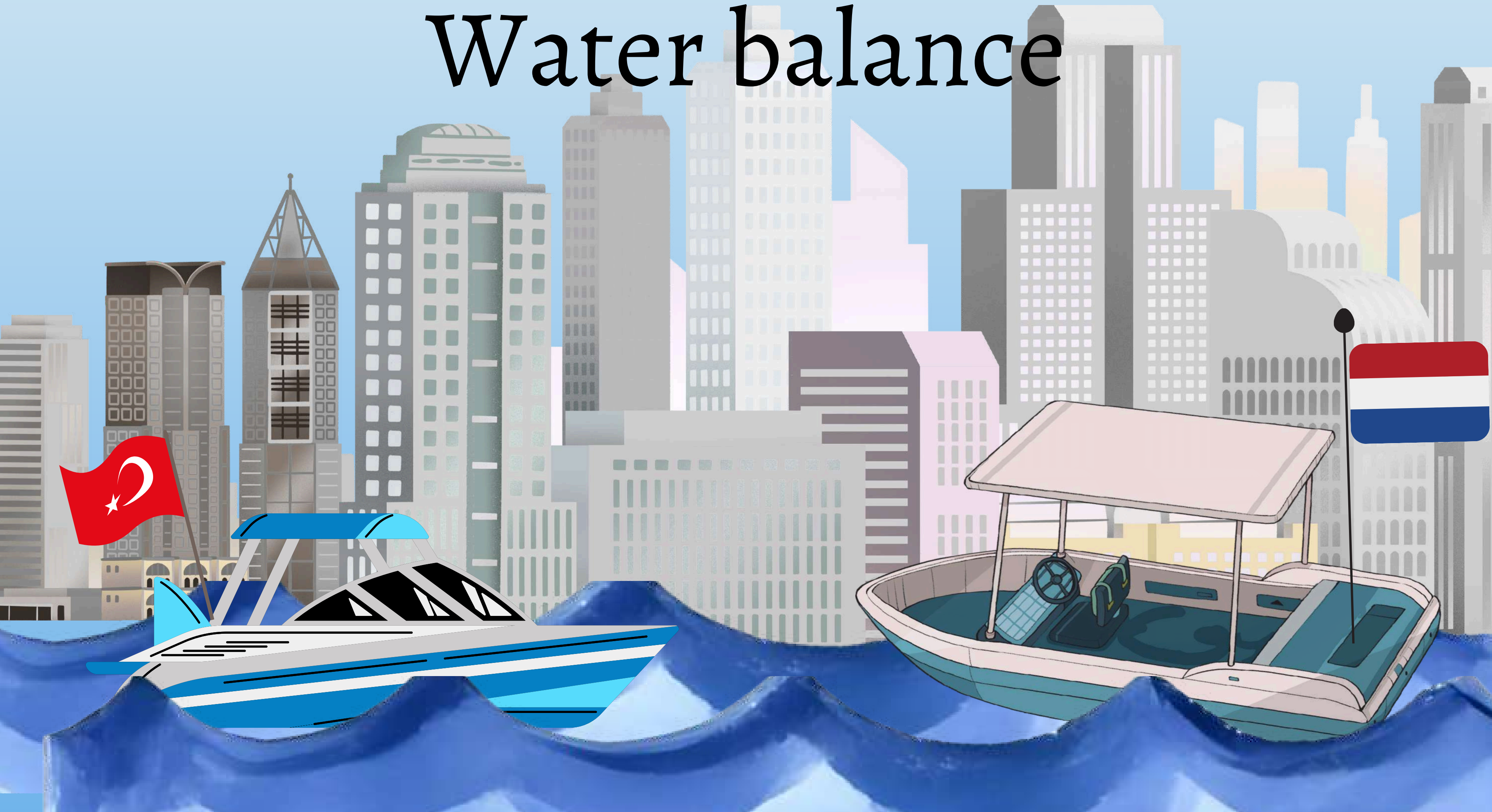


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Water balance



Istanbul

Some scientific studies suggest that large cities like Istanbul may face a risk of "water scarcity" or "Day Zero" (the day when water doesn't flow from the taps) by 2050. The negative impact of climate change on rainfall patterns, increasing water demand, and uncontrolled consumption all exacerbate this risk; therefore, it is believed that Istanbul's water resources could be depleted and a severe drought could occur.

On the other hand, water management policies, infrastructure improvements, water conservation, wastewater treatment, and recycling measures can improve Istanbul's future water situation. Successful practices can reduce the "risk of water scarcity."

Alphen

Alphen a/d Rijn is below sea level, about 0.5 metres.

Which means we could easily have a flood.

We are also near a river, the Rijn, which doesn't help.

With climate change the chance of flooding just increases and those (centi)metres climb up over time. We do have drains and a sewage system, however they'll probably become flooded quickly. This already happens when a lot of rain falls here...

Plus there aren't drains everywhere, so we often have puddles in the town.

Then we also have water scarcity during (extreme) heat, which is uncommon now, but could happen in the future thanks to climate change. It's important to address, because this impacts water supplies and the overall nature in the area.

Dutch people have a long history of fighting with water. We built dikes, mills that pump water away and built dams. For our solution, we suggest using the same things, only a bit modernized! Bigger, better and stronger dikes, dams and a modern version of the old mills that pump water. Plus a better drain and sewer system, capable of handling heavy rainfall.

We should also start thinking about storing water, so we always have enough water and never have a shortage.



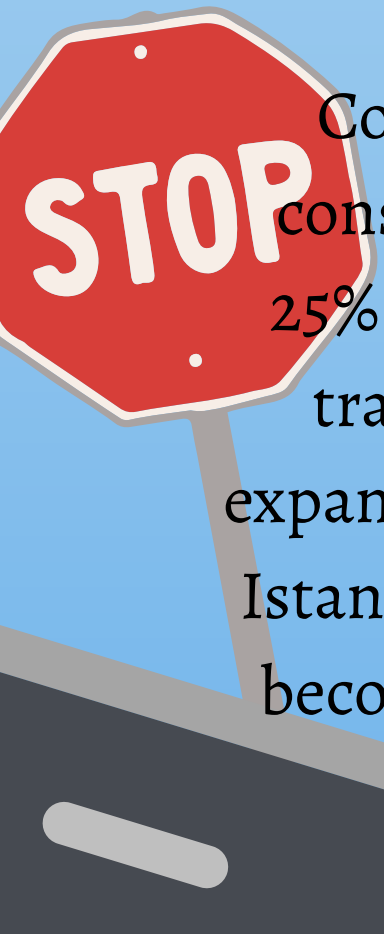
Urban planning



Istanbul

By 2050, urbanization in Istanbul is projected to have progressed towards a highly planned and sustainable structure. Thanks to urban transformation projects and smart city applications, the rate of well-planned urbanization is estimated to reach 75-80%. Renewals of buildings, particularly those renovated to mitigate earthquake risk, have played a significant role in this increase.

Conversely, the rate of unplanned and irregular construction is estimated to have decreased to 20-25%. This reduction has been achieved through the transformation of informal settlements and the expansion of green spaces. It can be said that by 2050, Istanbul will have made significant progress towards becoming a more livable, safe, and environmentally friendly city.



Alphen

The municipality of Alphen a/d Rijn has already been working on creating a more sustainable and functional city for some time now. They are doing so with malls and parks.

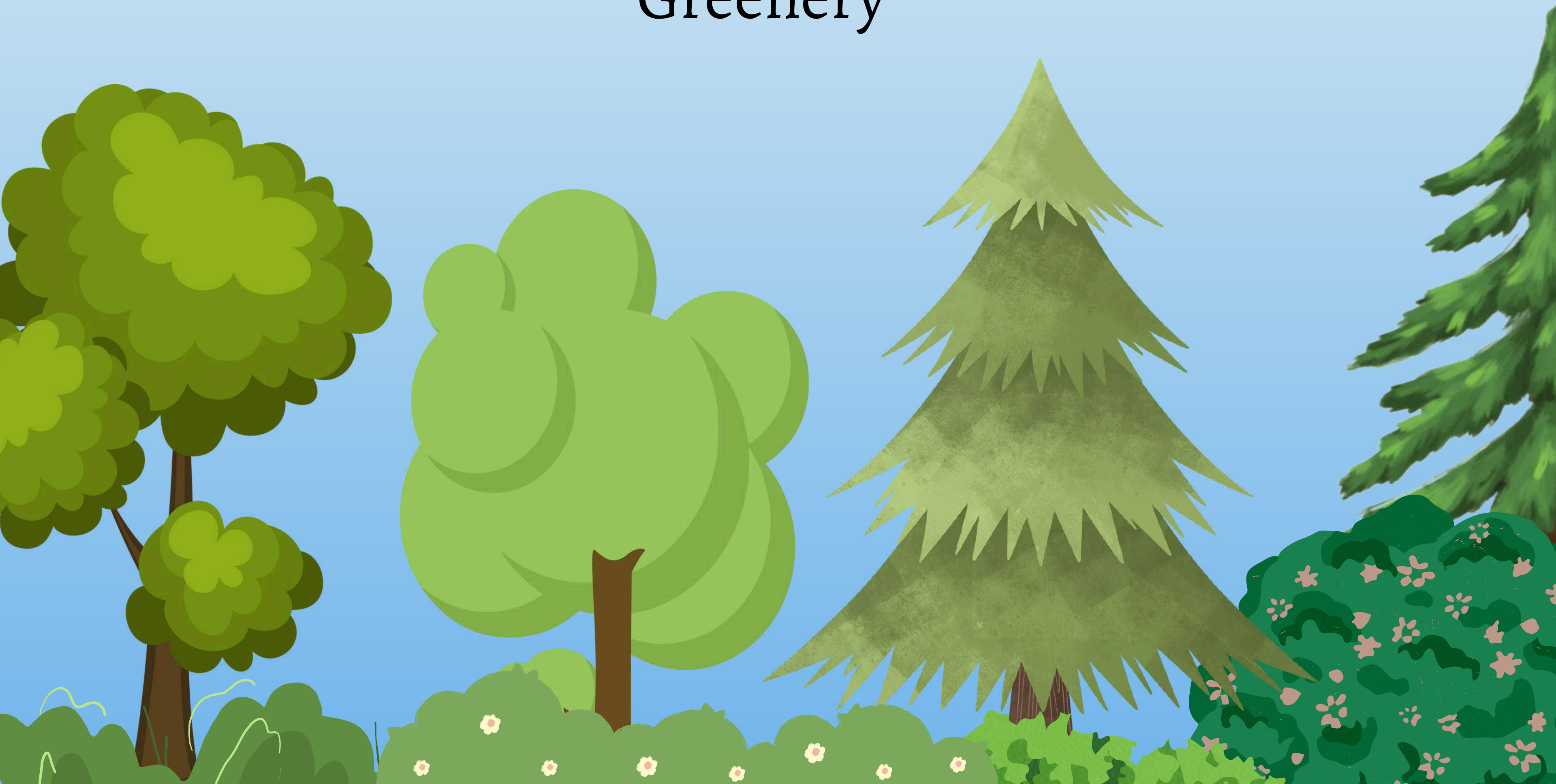
They are also creating a system more capable to deal with the water problem. This has already proven to have made Alphen a better city than it has been before.

But even after all this there are still problems. Some residents may have less facilities in their surroundings than others, while others may suffer from loud noises. City lay-out also is part of urban planning. In other cities streets may be straight and have 90° turns so it forms blocks. Alphen however has curved roads and uneven residential zones. That might be confusing for some.

Alphen's strategy of creating a better city is making zones whether large or small with necessities. They are doing so in residential areas. That makes the city seem vibrant and lively all throughout. This idea of urban planning has already been done by many cities. In 2025 we expect Alphen to have more of these mini market zones and we also expect 1 new large district. The district would include things like housing grocery stores and other necessities.



Greenery



Istanbul

By 2050, increasing green spaces in Istanbul has become a fundamental element in improving the quality of urban life.

Thanks to parks, urban forests, and coastal development projects, the percentage of green spaces throughout Istanbul is projected to reach 18-20%. With these developments, it is estimated that approximately 65% of the city's population will have access to a green area within walking distance.

However, due to Istanbul's dense population and limited space, inequalities in the distribution of green areas persist.

While the percentage of green spaces remains around 10-12% in the city center, this figure rises to 25-30% in new and planned residential areas. Overall, although the amount of green space in Istanbul will increase compared to previous years, protecting and equitably distributing these areas will remain a crucial issue in the future.

Alphen

In Alphen a/d Rijn there is quite a lot of greenery, only it's divided.

Most of the streets have some form of greenery, whether that is trees, bushes or some grass. That also means some areas have a lot of greenery while others may not. Which is a little problem.

Overall, there are 10 parks in Alphen a/d Rijn, varying in sizes.

In the future, there will probably be more greenery and not necessarily parks in Alphen. Alphen doesn't have a lot of room to expand, so building new parks probably won't happen that much.

The solution is to really add more greenery. Mostly in neighborhoods and on random spots. Parks should already have a lot of greenery, so they should instead be added in uncommon areas. For example rooftops of flats and other tall buildings. Another thing is that every area should have as much greenery as others.